Lucien Hennecart

CNRS Researcher | LAMFA, Universite de Picardie Jules Verne, Amiens

APPOINTMENTS

Since Dec 2024: Charge de recherche (CR), CNRS, LAMFA, Universite de Picardie Jules Verne, Amiens.

Sep 2021-Nov 2024: Postdoctoral researcher, School of Mathematics, University of Edinburgh; collaboration with Ben Davison.

Jan-May 2024: Uhlenbeck Postdoctoral Fellow, SLMath, Berkeley.

EDUCATION

2018-2021: PhD in mathematics, Universite Paris-Saclay (Orsay); advisor Olivier Schiffmann. Thesis defended on 13 July 2021.

2017-2018: M2 Arithmetique, Analyse, Geometrie, Paris-Saclay.

2017: Agregation de mathematiques (ranked 5th). 2014-2018: ENS de Rennes.

RESEARCH INTERESTS

Geometric representation theory; cohomological Hall algebras; BPS Lie algebras and invariants; moduli stacks and spaces; 2-Calabi-Yau categories; Donaldson-Thomas theory; nonabelian Hodge theory; quiver representations.

RESEARCH FUNDING

Principal investigator, ANR JCJC ReSCoS, 2027-2031 (EUR 260,710), French National Research Agency.

RECENT PUBLICATIONS AND PREPRINTS

2026: Commutativity of nilpotent cohomological Hall algebras of A^2. arXiv:2607.06512.

2026: Hecke operators on symplectic surfaces and chi-independence. arXiv:2607.01355.

2026: Degenerations of CoHAs of 2-Calabi-Yau categories. arXiv:2602.18102.

2025: The BPS decomposition theorem (with T. Kinjo). arXiv:2509.21298.

2025: Cohomological Mackey formula for quotient stacks. arXiv:2505.09483.

2024: Cohomological integrality for symmetric quotient stacks. arXiv:2408.15786.

2024: Cohomological integrality for weakly symmetric representations of reductive groups. arXiv:2406.09218.

2023: BPS algebras and generalised Kac-Moody algebras from 2-Calabi-Yau categories. arXiv:2303.12592.

2022: Geometric realisations of the unipotent enveloping algebra of a quiver. Adv. Math. 441 (2024).

SCIENTIFIC ORGANIZATION

GAT seminar, Amiens (since 2025); GRAAL meetings between Amiens and Lille (since 2026); Workshop on Geometric Representation Theory, Amiens (June 2026); formerly EGRET seminar in Edinburgh and events at SLMath and Skye.

TEACHING

Advanced M2 course on perverse sheaves in Paris (March-April 2027); mathematics teaching at Edinburgh and Orsay.

CONTACT AND LINKS

Lucien.Hennecart@u-picardie.fr | https://hennlu.github.io/ | https://arxiv.org/a/hennecart_l_1.html